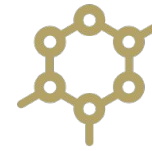


Effectiveness of Learning Modalities

Research conducted by Peter Trinh

Video Presentation Link: <https://www.youtube.com/watch?v=V9sOsrcSf9c>



INTRODUCTION

- Schools rely heavily on text based instruction, which may not match the best learning modality for all students
- Many studies show instructional modality affects comprehension and memory
- Cognitive Theory of Multimedia: Multimedia and active engagement reduces cognitive overload by distributing information and leads to deeper understanding
- Dual Coding Theory: Learning is strengthened when visual and verbal systems are used simultaneously
- This study investigates whether text, audio, or interactive lessons is the best modality for learning
- Research Question: How does instructional modality affect comprehension and memory retention?
- Hypothesis: Interactive Multimedia is the most effective modality.

DESIGN

- Participants: 10 individuals, 5 male, 5 female, high school graduate who has taken geometry before
- Structure: Each participant completed 3 lessons on area of 2D shapes, circumference and perimeter of shapes, and volume of 3D shape, each lesson was learned using a different modality: text only, audio only, and interactive multimedia
- Procedure:
 - Each participant spends 15 minutes learning a topic with a randomized modality.
 - They take a quiz on the subject
 - Repeat for all 3 types of modalities
 - After 24-48 hours, they take delayed recall quiz and complete a participant ratings survey on difficulty, engagement, and confidence of each modality
- Independent Variable: Instructional modality (text, audio, interactive)
- Dependent Variable: Participant quiz score, Delayed recall score, subjective ratings

RESULTS

Participant Quizzes Averages:

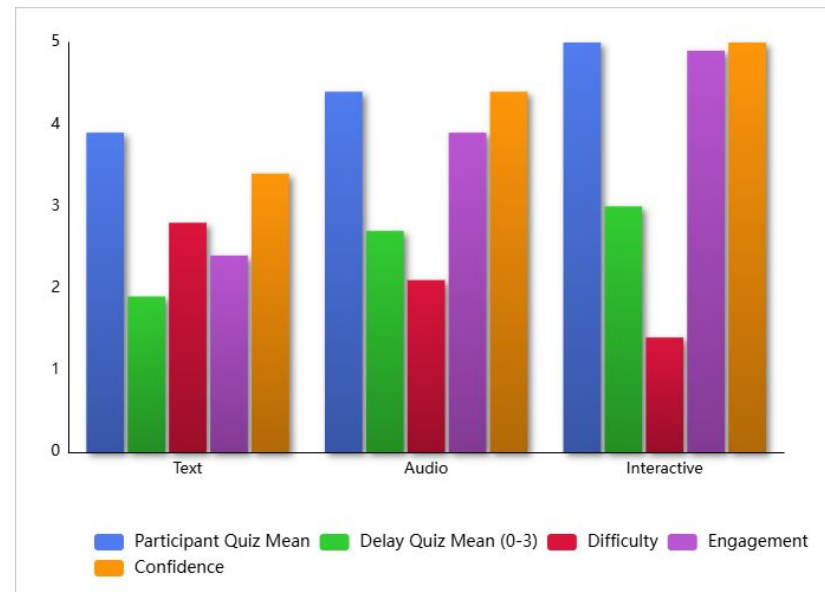
- Text: 3.9 / 5
- Audio: 4.4 / 5
- Interactive: 5 / 5

Delayed Recall Quizzes Averages:

- Text: 1.9 / 3
- Audio: Mean 4.4 / 5
- Interactive: 3 / 3

Survey Findings:

- Text based rated least engaging and most difficult
- Audio rated easier than text and slightly engaging
- Interactive rated easiest, most engaging, and highest level of confidence



CONCLUSION

- Learning modality effectiveness ranking:
 - 1st: Interactive
 - 2nd: Audio
 - 3rd: Text
- Interactive multimedia instruction produced the strongest learning outcome and retention
- Results support multiple theories such as: Cognitive Theory of Multimedia, Dual Coding Theory, and Cognitive Load Theory
- The experiment supports the need for modernization of education materials that still focus too heavily on only text based learning

Limitations

- Small sample size (N =10) reduces generalization and results may not reflect broader populations
- Simple geometry concepts were used for the learning process; effects may differ for more complex concepts
- Short quiz format may not fully measure conceptual understanding and retention

Future Plans/Applications

- Replicate with larger and more diverse samples
- Test more difficult and complex concepts
- Measure longer term memory retention across weeks instead of 1-2 days
- Apply findings to current education system to improve teaching technique effectiveness